

Unit 6: Practical Skills in Chemistry II

IA2 compulsory unit

Externally assessed

Unit description

Introduction

This unit consists of a written practical examination, covering the skills and techniques developed during practical work in Units 4 and 5, as well as the tests for anions and cations, gases and organic functional groups from Units 1 and 2.

Although the unit content contains eight core practical activities, the examination will not be limited to recall of these core practicals, there may be questions where students need to apply their knowledge to new practical situations.

Students should, therefore, develop their practical skills by completing a range of different practicals that require a variety of different techniques.

As students carry out practical activities, they should be encouraged to write laboratory reports using appropriate scientific, technical and mathematical language, conventions and symbols.

Development of practical skills, knowledge and understanding

Students are expected to develop experimental skills and knowledge and understanding of the necessary techniques by carrying out a range of practicals while they study Units 4 and 5.

This unit will assess students' knowledge and understanding of the practical procedures and techniques that they develop.

To prepare for assessment of this unit, centres should give students opportunities to carry out practical activities, to collect and analyse data, and to draw conclusions. Students should – at the least – carry out the eight core practicals in class. By completing these practicals, students will be able to:

- follow and interpret experimental instructions, covering the full range of laboratory exercises set throughout the course, with minimal help from the teacher
 - always work with interest and enthusiasm in the laboratory, completing most laboratory exercises in the time allocated
 - manipulate apparatus, use chemicals, carry out all common laboratory procedures and use data logging (where appropriate) with the highest level of skill that may be reasonably expected at this level
 - work sensibly and safely in the laboratory, paying due regard to health and safety requirements without the need for reminders from the teacher
 - gain accurate and consistent results in quantitative exercises, make the most of the expected observations in qualitative exercises and obtain products in preparations of high yield and purity.
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